CoRe-TFM: Selective Four-View Probability Reconciliation for Tabular Foundation Models

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Abstract

Tabular foundation models (TFMs) expose predictive probabilities that can be composed autoregressively for multivariate inference, but recent evidence shows that direct marginals and conditional predictions from the same frozen TFM need not correspond to a common joint distribution. We study post-hoc reconciliation of four views, $p(A \mid X)$, $p(B \mid X)$, $p(A \mid B, X)$, and $p(B \mid A, X)$, without retraining the underlying model. CoRe-TFM treats the two chain-rule joints as competing probabilistic views. Hard CoRe performs an information projection that exactly preserves direct marginals; Soft CoRe relaxes marginal preservation through KL penalties and admits generalized Sinkhorn-style updates; Selective CoRe uses held-out validation data to choose among unrepaired factorizations, pooling, and reconciliation. Controlled experiments with known conditional truth show that coherence, calibration, and probabilistic correctness are distinct. Exact-view perturbations identify when hard marginal preservation helps or hurts, while a heterogeneous task-mixture benchmark shows that Selective CoRe significantly outperforms the best fixed method and approaches a non-deployable candidate-family oracle. A five-fold TabICLv2 pilot on UCI Car Evaluation independently confirms substantial cross-query incompatibility and shows that the best unreconciled chain direction can still outperform both hard and soft reconciliation. The results support selective rather than unconditional reconciliation; a broader released-model matrix remains necessary before making average TFM performance claims.

Keywords: tabular foundation models, probabilistic consistency, probability reconciliation, proper scoring rules, information projection

1 Introduction

Tabular foundation models have rapidly become strong general-purpose predictors for structured data. TabPFN established that a transformer pretrained on synthetic tasks can perform competitive tabular prediction by in-context inference (Hollmann et al., 2025). More recent systems such as TabPFN-3 (Grinsztajn et al., 2026), TabICLv2 (Qu et al., 2026), and TabFM (Google Research, 2026) extend this paradigm in scale, speed, and task coverage. Their predictive probabilities are attractive not only for classification but also for uncertainty-sensitive decisions, conditional queries, sequential simulation, imputation, and multivariate scenario generation.

These uses expose a structural question that standard single-target benchmarks do not test. Suppose a frozen predictor is queried for two categorical targets A and B. It can produce direct marginals $p(A \mid X)$ and $p(B \mid X)$, together with conditional predictions $p(A \mid B, X)$ and $p(B \mid A, X)$. If all four were views of one joint distribution, marginalization would recover the direct predictions and the two chain-rule factorizations would agree.

Kloetergens et al. (2026) recently tested these requirements and found systematic marginalization and factorization violations across evaluated TFMs and datasets. That result establishes a diagnosis: probabilities produced by the same frozen TFM need not be mutually representable by one joint. It does not determine the appropriate repair. Constructing any single normalized joint makes its own marginals and conditionals compatible by definition, so zero post-repair inconsistency is not evidence that the repaired probabilities are useful. A coherent distribution can have worse proper scores, worse calibration, or distorted dependence.

The distinction matters because recent TFM work increasingly evaluates the full predictive distribution. ScoringBench shows that model rankings depend on the proper scoring rule used (Landsgesell and Knoll, 2026); conditional-density and uncertainty studies show strong predictive performance alongside reliability trade-offs (Izbicki and Rodrigues, 2026; Johnson et al., 2026); and DistPFN demonstrates that targeted test-time posterior adjustment can improve a distinct failure mode, label shift, without retraining (Lee et al., 2026). CoRe-TFM addresses neither generic calibration nor label shift. It addresses incompatibility among simultaneous probability views produced by the same predictor.

We formulate reconciliation as information projection over a finite categorical joint. The two autoregressive factorizations are treated as competing joint views. Arithmetic and geometric pooling provide principled KL-barycenter baselines. Hard CoRe projects a consensus distribution onto the transportation polytope defined by the two direct marginals. Soft CoRe relaxes exact marginal preservation with KL penalties and can be solved by generalized matrix scaling. Selective CoRe treats repair as a model-selection problem: validation data may select an original factorization, a simple pool, or a reconciliation policy.

The empirical design is diagnostic rather than winner-takes-all. First, a surrogate multiclass data-generating process provides exact $P^*(A, B \mid X)$ while finite-data estimators induce realistic coupled prediction errors. Second, an exact-view perturbation benchmark starts from the exact joint and independently perturbs the two marginals and two conditional families, isolating the reliability assumptions encoded by hard and soft reconciliation. Third, a heterogeneous task-mixture benchmark samples the reliability of all four views independently and evaluates whether Selective CoRe adapts to the task. A validation-size study measures how selection regret changes as held-out evidence grows.

Our contributions are:

1. a four-view formulation of TFM probability reconciliation that separates compatibility from calibration and predictive correctness;

2. Hard and Soft CoRe objectives, including a four-view KL decomposition and generalized Sinkhorn-style updates;

3. a Selective CoRe policy with a finite-candidate validation-selection guarantee;

4. an exact-view mechanism-identification benchmark showing when marginal preservation is beneficial or harmful;

5. a heterogeneous 60-task benchmark in which Selective CoRe significantly outperforms the best fixed method and remains close to a full candidate-family oracle;

6. a five-fold released-model pilot on TabICLv2 and UCI Car Evaluation that reproduces the qualitative magnitude of inconsistency while showing that unconditional reconciliation need not improve joint log loss; and

7. a reproducible evaluation protocol covering joint, marginal, conditional, calibration, truth-distance, dependence-fidelity, and computational metrics.

2 Related Work

2.1 Tabular foundation models and probabilistic prediction

TabPFN introduced prior-data fitted networks as pretrained Bayesian-like tabular predictors and demonstrated strong small-data performance (Hollmann et al., 2025). TabPFN-3 expands this approach to much larger tables and test-time compute scaling (Grinsztajn et al., 2026). TabICLv2 provides an open, scalable in-context model with built-in handling of heterogeneous tabular features (Qu et al., 2026). TabFM is another recent zero-shot TFM release (Google Research, 2026). These systems differ in pretraining priors, scaling strategy, checkpoint access, and inference behavior, making a model-agnostic post-hoc consistency layer attractive.

Probability quality is now a first-class evaluation target. Proper-scoring benchmarks can reorder TFM conclusions (Landsgesell and Knoll, 2026); conditional-density evaluation suggests that foundation models can be strong distributional predictors while still exhibiting calibration trade-offs (Izbicki and Rodrigues, 2026); and uncertainty studies emphasize that accurate means or classes need not imply reliable predictive uncertainty (Johnson et al., 2026). DistPFN (Lee et al., 2026) is especially relevant as evidence that post-hoc probability adjustment for frozen TFMs can be useful, but it corrects label-shift bias rather than cross-query incompatibility.

2.2 Consistency of conditional distributions

The closest diagnostic work is Kloetergens et al. (2026), which formalizes marginalization and factorization tests for TFMs. A broader theoretical precursor is Majid et al. (2025), who study when families of learned conditional distributions can be recovered from a consistent joint and derive path, autoregressive, and swap consistency conditions. Compatibility of separately specified conditionals is older still; Mure (2017) studies compromise distributions induced by incompatible conditionals. Conditioning consistency also appears in conditional neural processes (Young, 2026). CoRe-TFM therefore does not claim to discover conditional inconsistency. Its narrower contribution is a post-hoc reconciliation family for the four views naturally exposed by frozen TFMs, evaluated by predictive fidelity rather than consistency alone.

2.3 Joint tabular modeling as an alternative

A different response to multivariate tabular inference is to train a model that represents a joint distribution directly. TabNAT learns order-agnostic conditional generation over mixed-type tables (Zhang et al., 2025), while TabDiff uses a mixed-type diffusion process for joint generation and imputation (Shi et al., 2025). Such approaches avoid some composition problems by construction but require a generative training objective and model family. CoRe-TFM asks a complementary question: when a strong frozen discriminative TFM already exists, can its incompatible probability queries be reconciled after inference?

2.4 Information projection and generalized scaling

Hard reconciliation is an information projection onto a transportation polytope, closely related to iterative proportional fitting and Bregman projections (Benamou et al., 2015). Sinkhorn scaling is a central computational primitive for entropic transport (Cuturi, 2013), and relaxed marginal constraints lead to generalized scaling updates (Chizat et al., 2018). We use these established tools as optimization machinery. Algorithmic novelty is not claimed for Sinkhorn scaling itself.

3 Problem Formulation

For a context data set D, test covariates x, and categorical targets $A \in \{1, \dots, K_A\}$ and $B \in \{1, \dots, K_B\}$, query the frozen predictor four ways:

$$p_A(a) = \hat{p}(A = a \mid x, D), \qquad p_B(b) = \hat{p}(B = b \mid x, D), \tag{1}$$

$$p_{A|B}(a \mid b) = \hat{p}(A = a \mid B = b, x, D), \qquad p_{B|A}(b \mid a) = \hat{p}(B = b \mid A = a, x, D). \tag{2}$$

These induce two normalized joints,

$$J^{B\to A}(a, b) = p_B(b)p_{A|B}(a \mid b), \tag{3}$$

$$J^{A\to B}(a, b) = p_A(a)p_{B|A}(b \mid a). \tag{4}$$

A coherent system satisfies $J^{B\to A} = J^{A\to B}$. We measure factorization inconsistency by

$$\mathrm{TV}(J^{B\to A}, J^{A\to B}) = \frac{1}{2} \sum_{a,b} |J^{B\to A}(a, b) - J^{A\to B}(a, b)|. \tag{5}$$

Each direction also contains one non-trivial marginalization check because the first factor's marginal is satisfied by construction while the other need not match its direct prediction.

Our target is not merely any coherent Q. We seek a distribution that trades off agreement with $J^{B\to A}$ and $J^{A\to B}$, fidelity to p_A and p_B, predictive scoring, and dependence preservation. On real data, the true posterior is unknown, so proper scores and calibration are primary. On controlled synthetic tasks, exact P^* permits direct truth-distance evaluation.

4 CoRe-TFM

4.1 KL-barycenter baselines

For $w \in [0, 1]$, define the arithmetic and geometric pools

$$Q_{\mathrm{arith},w} = wJ^{B\to A} + (1 - w)J^{A\to B}, \tag{6}$$

$$M_w(a, b) = \frac{J^{B\to A}(a, b)^w J^{A\to B}(a, b)^{1-w}}{\sum_{a',b'} J^{B\to A}(a', b')^w J^{A\to B}(a', b')^{1-w}}. \tag{7}$$

The arithmetic pool minimizes $w\,\mathrm{KL}(J^{B\to A}\|Q) + (1-w)\,\mathrm{KL}(J^{A\to B}\|Q)$, whereas M_w minimizes $w\,\mathrm{KL}(Q\|J^{B\to A}) + (1-w)\,\mathrm{KL}(Q\|J^{A\to B})$. This makes both strong baselines rather than arbitrary averaging rules.

4.2 Hard CoRe: marginal-preserving reconciliation

Let

$$\mathcal{T}(p_A, p_B) = \left\{ Q \geq 0 : \sum_b Q(a,b) = p_A(a),\ \sum_a Q(a,b) = p_B(b) \right\}. \tag{8}$$

Hard CoRe is

$$Q_{\mathrm{hard}} = \arg \min_{Q \in \mathcal{T}(p_A, p_B)} \mathrm{KL}(Q\|M_w). \tag{9}$$

Proposition 1 (Feasibility) *For any categorical marginals p_A and p_B, $\mathcal{T}(p_A, p_B)$ is non-empty.*

Proof The independent joint $Q_0(a,b) = p_A(a)p_B(b)$ has exactly the required row and column sums. ∎

Proposition 2 (Uniqueness under positive support) *If $M_w(a,b) > 0$ for every cell, the Hard CoRe objective has a unique optimum on $\mathcal{T}(p_A, p_B)$.*

Proof The feasible set is convex and compact. $\mathrm{KL}(Q\|M_w)$ is strictly convex in Q on the positive-support simplex, so a feasible minimizer exists and is unique. ∎

Iterative proportional fitting solves the finite problem by alternating row and column scaling. Hard CoRe changes dependence while exactly preserving the two direct marginals, which is useful only when those marginals deserve special trust.

4.3 Soft CoRe: four-view reconciliation

Hard marginal preservation can preserve a bad marginal. We therefore define

$$\begin{aligned} Q_{\mathrm{soft}} = \arg\min_Q\ & w\,\mathrm{KL}(Q\|J^{B\to A}) + (1-w)\,\mathrm{KL}(Q\|J^{A\to B}) \\ & + \lambda_A\,\mathrm{KL}(Q_A\|p_A) + \lambda_B\,\mathrm{KL}(Q_B\|p_B). \end{aligned} \tag{10}$$

The first two terms equal $\mathrm{KL}(Q\|M_w)$ up to constants independent of Q. More importantly, the KL chain rule gives

$$\mathrm{KL}(Q\|J^{B\to A}) = \mathrm{KL}(Q_B\|p_B) + \mathbb{E}_{b\sim Q_B}\,\mathrm{KL}(Q_{A|B=b}\|p_{A|B=b}), \tag{11}$$

$$\mathrm{KL}(Q\|J^{A\to B}) = \mathrm{KL}(Q_A\|p_A) + \mathbb{E}_{a\sim Q_A}\,\mathrm{KL}(Q_{B|A=a}\|p_{B|A=a}). \tag{12}$$

Thus Eq. (10) is genuinely a four-view objective: both marginals and both conditional families influence the solution.

4.4 Generalized matrix scaling

For positive M_w, the optimum has scaling form $Q \propto \mathrm{diag}(u) M_w \mathrm{diag}(v)$. Let

$$\tau_A = \frac{\lambda_A}{1+\lambda_A}, \qquad \tau_B = \frac{\lambda_B}{1+\lambda_B}. \tag{13}$$

Alternating updates are

$$u \leftarrow \left(\frac{p_A}{M_w v}\right)^{\tau_A}, \tag{14}$$

$$v \leftarrow \left(\frac{p_B}{M_w^\top u}\right)^{\tau_B}. \tag{15}$$

These are generalized Sinkhorn-style updates for KL-relaxed marginals (Chizat et al., 2018). Our implementation retains an independent exponentiated-gradient solver and generic constrained optimizer on small instances as numerical cross-checks.

4.5 Selective CoRe

A single reconciliation strength cannot be optimal if view reliability changes by task. Selective CoRe therefore searches a finite policy family on an inner validation split. The current family contains the two original orders, arithmetic pooling, weighted geometric pooling, weighted Hard CoRe, and weighted Soft CoRe across five direction weights and seven marginal penalties, for 48 distinct candidates after removing duplicates. The chosen policy is then frozen and evaluated on held-out test data.

Let $L(Q)$ denote population joint log loss and $\hat{L}_v(Q)$ validation loss on n_v independent examples. To make the selection guarantee finite, probabilities are floored at ϵ so loss is bounded by $B = -\log \epsilon$.

Proposition 3 (Finite-candidate selection bound) *Let $\mathcal{C}$ contain M candidate distributions and let $\hat{Q} = \arg\min_{Q \in \mathcal{C}} \hat{L}_v(Q)$. With probability at least $1-\delta$,*

$$L(\hat{Q}) \le \min_{Q \in \mathcal{C}} L(Q) + 2B\sqrt{\frac{\log(2M/\delta)}{2n_v}}. \tag{16}$$

Proof Hoeffding's inequality and a union bound give uniform deviation at most $c = B\sqrt{\log(2M/\delta)/(2n_v)}$ over $\mathcal{C}$. If Q^* minimizes population loss in $\mathcal{C}$, then $L(\hat{Q}) \le \hat{L}_v(\hat{Q}) + c \le \hat{L}_v(Q^*) + c \le L(Q^*) + 2c$. ■

The bound is intentionally conservative because B can be large, but it makes one qualitative prediction that we test directly: selection regret should decline with validation sample size.

5 Experimental Design

5.1 Metrics and statistical principles

Because the output is a probability distribution, proper scores are primary (Gneiting and Raftery, 2007). We report joint NLL and Brier score; marginal NLL/Brier/accuracy; NLL and Brier for both conditionals induced by a candidate joint; factorization and marginalization TV; marginal and reconciliation distortion; mutual-information error; and runtime. Expected calibration error is retained as a descriptive diagnostic because it depends on binning and is not itself a proper scoring rule (Guo et al., 2017).

When exact $P^*(A, B \mid X)$ is known, we report exact expected log loss

$$\mathcal{L}_{P^*}(Q) = -\mathbb{E}_X \sum_{a,b} P^*(a, b \mid X) \log Q(a, b \mid X), \tag{17}$$

plus TV and Jensen–Shannon distance to truth. Expected scores remove test-label Monte Carlo noise.

All adaptive decisions use validation labels disjoint from test observations. Paired effects are computed within random seed or task. Prespecified comparisons use two-sided Wilcoxon signed-rank tests; multiple comparisons within the exact-view regimes use Holm correction. Effect sizes accompany p-values.

5.2 Surrogate DGP benchmark

A multiclass DGP provides exact P^*. Gaussian features generate $A \mid X$ by softmax, while $B \mid A, X$ follows a second softmax with dependence parameter γ. Smooth nonlinear terms create misspecification for a multinomial logistic-regression surrogate. We vary one factor at a time: $\gamma \in \{0, 0.5, 1.5, 3\}$; $n \in \{250, 1000, 5000\}$; $d \in \{5, 20, 50\}$; target cardinality in $\{2, 3, 5\}$; and three imbalance levels. Each condition uses 10 independent seeds.

5.3 Exact-view perturbation benchmark

The surrogate benchmark couples error across all four queries. We therefore start from exact P^*, derive compatible p^*_A, p^*_B, $p^*_{A|B}$, and $p^*_{B|A}$, and perturb each in log-probability space before renormalization. Six regimes are evaluated across 20 paired seeds: uniform low noise; reliable marginals; reliable conditionals; reliable $B \to A$ factorization; reliable $A \to B$ factorization; and uniformly high noise. This benchmark identifies which assumption makes a reconciliation method appropriate.

5.4 Heterogeneous task-mixture benchmark

To test Selective CoRe, we generate 60 independent tasks. On each task, the noise scale of each of the four views is drawn independently from a log-uniform distribution on $[0.04, 0.75]$. Each task has 1,500 observations and a three-by-three target joint; 30% of observations are used for validation and the remainder for exact expected test scoring. Selective CoRe searches the 48-candidate family described above. For analysis only, we also compute a non-deployable full-family oracle using exact test truth.

5.5 Validation-size sensitivity

A second mixture study uses 30 independently perturbed tasks with 2,600 observations each. A common test set is held fixed while selection is repeated with nested validation sizes $n_v \in \{100, 250, 500, 800\}$. This directly evaluates the qualitative prediction of Proposition 3.

Table 1: Best mean exact expected NLL in each exact-view perturbation regime (20 paired seeds).

Regime	Best method	Expected NLL
Uniform low noise	Geometric pool	1.452772
Direct marginals reliable	Soft CoRe ($\lambda = 10$)	1.460180
Conditionals reliable	Arithmetic pool	1.481462
$B \to A$ reliable	$J^{B \to A}$	1.448507
$A \to B$ reliable	$J^{A \to B}$	1.449996
All four views noisy	Arithmetic pool	1.512490

6 Controlled Results

6.1 Coherence alone is insufficient

The surrogate benchmark rejects a universal-ranking interpretation. At strong dependence ($\gamma = 3$), Adaptive Soft CoRe obtains mean joint NLL 1.5854, compared with 1.6082 for arithmetic pooling and 1.6048 for Hard CoRe. A validation-selected original order obtains 1.5865, showing that adaptive repair is competitive but that selecting a reliable factorization can capture most of the available signal. The independent joint is much worse at 1.7435 despite being perfectly coherent and exactly preserving both direct marginals. Coherence is therefore a constraint, not an accuracy metric.

At $n = 250$, arithmetic pooling is stronger than adaptive repair, while at larger sample sizes weighted pooling and selective policies become competitive. Raw inconsistency magnitude also fails as a universal repair trigger: across the dependence sweep, factorization TV has negligible rank association with Adaptive Soft's dividend over a validation-selected original order ($\rho \approx 0.065$), but a stronger association with its advantage over unweighted arithmetic pooling ($\rho \approx 0.422$, $p \approx 0.0067$). Larger inconsistency can indicate that symmetric averaging is inadequate without proving that repair is better than trusting one order.

6.2 Exact-view perturbations identify the reliability assumption

Table 1 and Fig. 1 summarize the mechanism study. The best method changes exactly when the trust structure changes.

When direct marginals are reliable and conditionals are noisy, Hard CoRe improves over arithmetic pooling by 0.0138 NLL on average; the Holm-adjusted paired Wilcoxon p-value is 8×10^{-6}. Strong Soft CoRe is slightly better still. When the direct marginals are noisy and conditionals reliable, the conclusion reverses: Hard CoRe is worse than arithmetic by 0.1423 NLL on average ($p_{\text{Holm}} = 8 \times 10^{-6}$). Adaptive Soft is 0.1388 better than Hard CoRe but remains slightly worse than arithmetic. Exact marginal preservation can therefore amplify a bad marginal forecast.

Directional regimes provide a third sanity check. When the $B \to A$ factorization is reliable, $J^{B \to A}$ is best and Adaptive Soft is only 0.00054 NLL worse than the validation-selected order. The analogous pattern holds for $J^{A \to B}$. Under symmetric low or high noise, arithmetic/geometric pooling is difficult to beat. A reconciliation algorithm cannot create

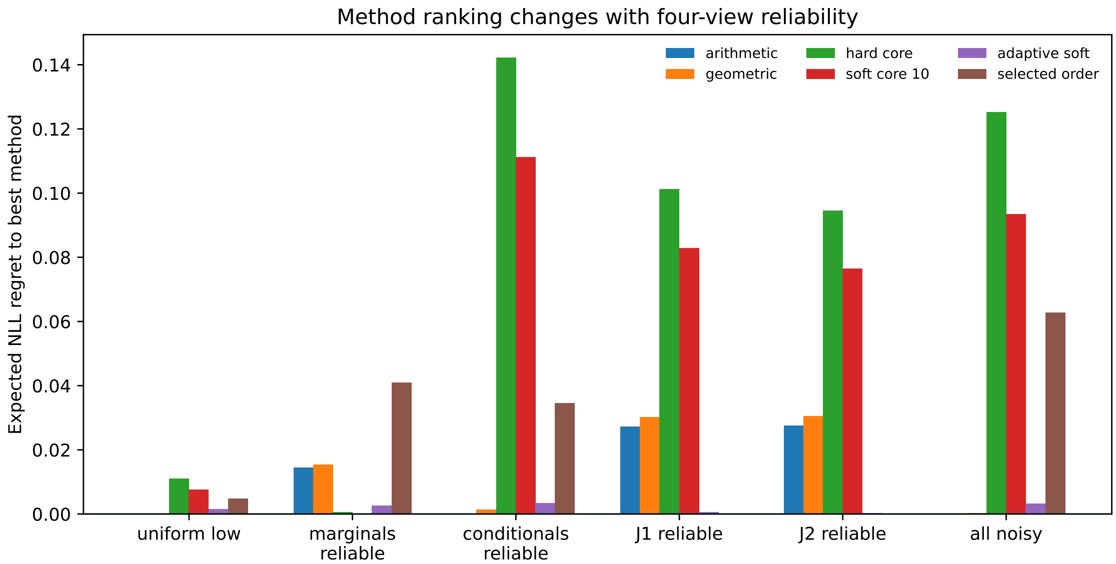

Figure 1: Expected NLL regret to the best method in each exact-view reliability regime. Hard preservation is useful when direct marginals are reliable and costly when they are not.

information absent from its inputs, so a correct deployment policy must be able to retain a trustworthy direction or simple pool.

6.3 Selective CoRe adapts across heterogeneous tasks

The heterogeneous task-mixture study is the strongest direct test of Selective CoRe. Table 2 reports exact expected NLL over 60 tasks. Selective CoRe has the best mean score among deployable methods, 1.49838, compared with 1.50419 for the best fixed method, arithmetic pooling. The paired difference is -0.00581 NLL; Selective CoRe wins on 41/60 tasks and the Wilcoxon p-value is 5.79×10^{-7}.

The policy does not collapse to one candidate. Across 60 tasks it selects Soft CoRe 22 times, geometric pooling 21 times, an unrepaired original order 10 times, arithmetic pooling 6 times, and Hard CoRe once. This diversity is expected under independently sampled view reliability and is evidence that the selector is adapting rather than merely tuning one global hyperparameter. Fig. 2 shows the paired task-level gain over arithmetic pooling.

6.4 More validation data reduces selection regret

The validation-size study supports the finite-candidate analysis. Mean regret to the full candidate-family oracle falls monotonically from 0.00464 at 100 validation observations to 0.00099 at 800 (Table 3). The mean gain over arithmetic grows from 0.00274 to 0.00639. The finite-sample bound is not numerically tight, but its $n_v^{-1/2}$ dependence captures the observed direction of improvement.

Table 2: Heterogeneous 60-task mixture benchmark. Oracle regret is measured against the same full 48-candidate family available to selection, evaluated with exact test truth only for analysis.

Method	Mean NLL	SE	Mean full-oracle regret
Selective CoRe	**1.498381**	0.007410	**0.001317**
Arithmetic	1.504194	0.007498	0.007130
Geometric	1.505179	0.007532	0.008114
Soft CoRe ($\lambda = 1$)	1.514203	0.008111	0.017138
$J^{A\to B}$	1.516449	0.009160	0.019384
$J^{B\to A}$	1.532221	0.009007	0.035156
Hard CoRe	1.548322	0.011195	0.051257

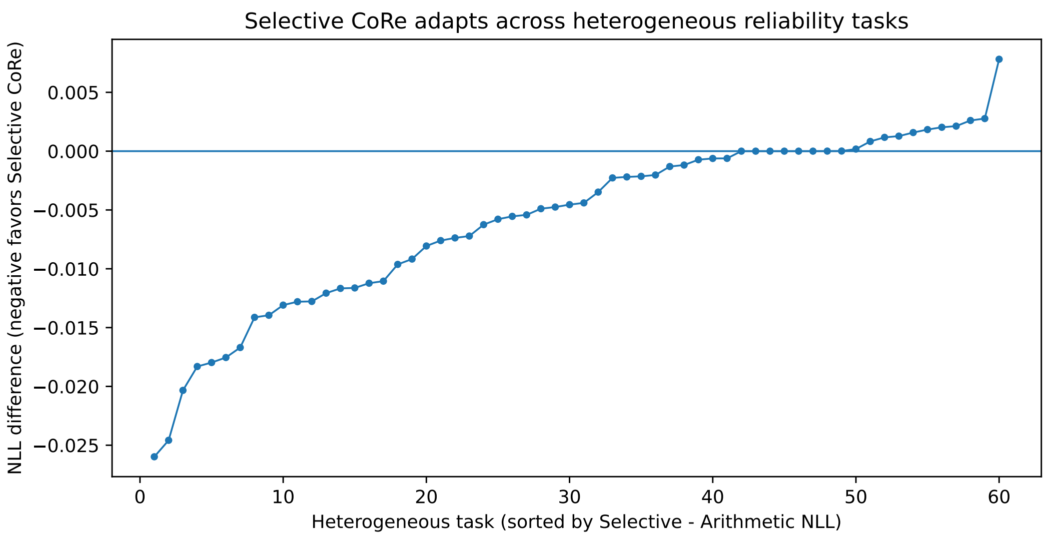

Figure 2: Task-level difference in exact expected NLL between Selective CoRe and arithmetic pooling, sorted across the 60 heterogeneous reliability tasks. Negative values favor Selective CoRe.

6.5 Directional defects are not correctness signals

A natural label-free heuristic is to trust the factorization whose implied missing marginal is closer to its direct prediction. We tested this rule in the surrogate DGP. Direction-selection accuracy weakens toward or below chance as dependence increases. We retain this as a negative ablation: a small internal consistency defect is not evidence that a factorization is closer to P^*.

Table 3: Validation-size sensitivity across 30 heterogeneous tasks.

n_v	Mean Selective NLL	Mean oracle regret	Mean gain over arithmetic
100	1.485699	0.004639	0.002742
250	1.483242	0.002182	0.005199
500	1.482888	0.001827	0.005553
800	1.482054	0.000994	0.006387

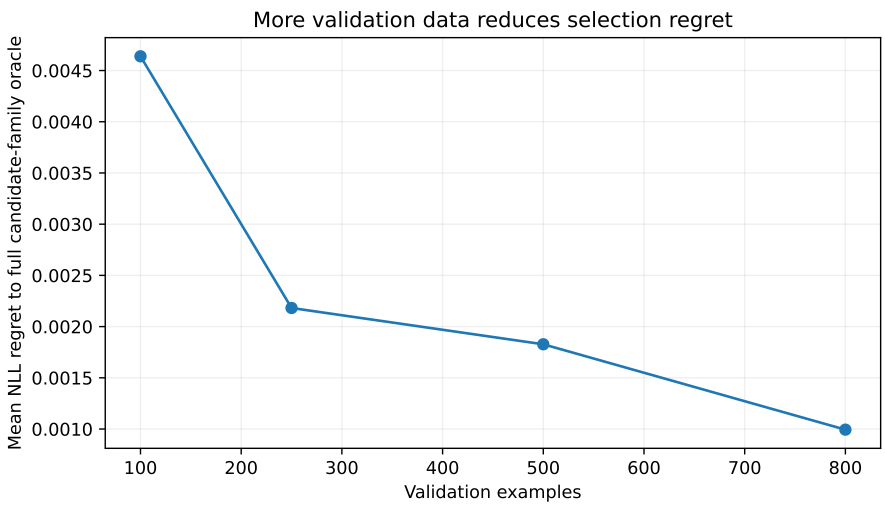

Figure 3: Mean exact NLL regret of Selective CoRe to the full candidate-family oracle as validation sample size increases.

6.6 Computational overhead

On a local batch of 5,000 5×5 joints, generalized scaling reaches the same soft objective as an independent exponentiated-gradient solver while requiring approximately 0.022 seconds versus 0.101 seconds. The intended interpretation is practical: reconciliation is lightweight relative to TFM inference. We do not claim optimization novelty over generalized Sinkhorn methods.

7 Real-TFM Pilot and Benchmark Protocol

The controlled evidence establishes how reconciliation behaves when probability-view reliability can be measured or manipulated. We now add a first released-model pilot, while keeping the broader model–data set matrix pre-specified rather than claiming that one cell establishes average TFM gains.

Table 4: Five-fold TabICLv2 pilot on UCI Car Evaluation. Values are fold mean ± sample standard deviation. Lower is better for both metrics. This CPU pilot uses one estimator and is not an exact reproduction of prior published settings.

Method	Joint NLL	Marginal distortion
$B \to A$ ($J^{B\to A}$)	1.12089 ± 0.00471	0.02495 ± 0.00294
$A \to B$ ($J^{A\to B}$)	1.18400 ± 0.00922	0.02162 ± 0.00118
Independent	1.55117 ± 0.01643	0.00000 ± 0.00000
Arithmetic pool	1.14625 ± 0.00559	0.02329 ± 0.00205
Geometric pool	1.13434 ± 0.00786	0.02762 ± 0.00177
Hard CoRe	1.20412 ± 0.02819	$< 2 \times 10^{-13}$
Soft CoRe ($\lambda = 1$)	1.13840 ± 0.00982	0.02253 ± 0.00183

7.1 Five-fold TabICLv2 pilot on Car Evaluation

We evaluate TabICLv2 on UCI Car Evaluation, using $A =$ class and $B =$ safety, with the remaining five categorical attributes as X (Bohanec, 1988). The data contain 1,728 rows. We use five-fold stratified cross-validation on A with shuffle enabled and random seed 42. For CPU tractability, the pilot uses one TabICLv2 estimator, key–value caching, and the released 2026 checkpoint. This is therefore not an exact reproduction of Kloetergens et al. (2026): their fold seed is not published and their default inference configuration differs.

Across the five folds, mean factorization inconsistency is 0.06931 ± 0.00354 TV. The corresponding value reported by Kloetergens et al. (2026) for TabICLv2 on Car is 0.0644 ± 0.0058. Given the non-identical split and inference configuration, we treat this only as a qualitative reproduction of the inconsistency scale, not as a numerical replication.

Table 4 reports joint NLL and marginal distortion. The key result is negative but informative: the $B \to A$ factorization is the best mean-NLL method on every aggregate comparison, at 1.12089 ± 0.00471. Geometric pooling reaches 1.13434 ± 0.00786, Soft CoRe with $\lambda_A = \lambda_B = 1$ reaches 1.13840 ± 0.00982, and Hard CoRe reaches 1.20412 ± 0.02819. Thus the released model exhibits clear incompatibility, yet enforcing a single repaired joint does not improve the best raw factorization. Hard CoRe nearly eliminates marginal distortion by construction, but that structural success coincides with worse predictive score. This is exactly the distinction between compatibility and correctness that motivates Selective CoRe.

7.2 Pre-specified broader benchmark

The pre-specified model set remains TabPFN-3 (Grinsztajn et al., 2026), TabICLv2 (Qu et al., 2026), and TabFM (Google Research, 2026). The initial data set list follows the categorical target pairs of Kloetergens et al. (2026): Anneal, Credit, Phishing, MIC, Customer, Car, Marketing, Wine, Nursery, and Diamonds. Each outer fold must extract all four probability views; an inner split chooses direction weight, marginal penalty, or Selective CoRe policy; no test information may influence selection. Cross-dataset comparisons use data set–model cells as statistical units rather than treating individual rows as independent replications.

The pilot is sufficient to demonstrate that the implemented four-view extraction and reconciliation pipeline operates on a released TFM, but it is not sufficient for an average-performance claim. Until the broader matrix is complete, this document remains a working draft rather than a submission-ready manuscript.

8 Discussion

8.1 Coherence, calibration, and correctness are different properties

Compatibility asks whether multiple probability queries can be represented by one joint. Calibration asks whether stated probabilities match empirical frequencies. Proper scoring evaluates the complete predictive distribution. Distance to P^* asks whether a controlled posterior is recovered. The experiments show that these properties can disagree. The independent baseline is coherent and can be poor; Hard CoRe can preserve two accurate marginals or two inaccurate marginals with equal mathematical success; and a reliable raw factorization can outperform a more coherent repair.

8.2 Marginal penalties have a reliability interpretation

The exact-view study gives λ a statistical meaning. Large λ expresses trust in direct marginals, while small λ allows conditional evidence to move them. The best penalty is therefore not merely a numerical tuning constant. It is a task-specific estimate of relative reliability among four model queries.

8.3 Why selection is the practical endpoint

Released TFMs differ in pretraining prior, architecture, scale, and inference strategy. Relative quality of direct and conditional queries can plausibly vary by model and data set. A universal hard constraint is therefore poorly matched to the setting. Selective CoRe operationalizes that uncertainty by keeping original factorizations and simple pooling inside the policy class. The task-mixture experiment shows why this flexibility matters.

8.4 Relation to joint generative models

Joint tabular generators such as TabNAT and TabDiff avoid some composition problems by directly modeling mixed-type joint distributions (Zhang et al., 2025; Shi et al., 2025). That is a compelling alternative when retraining or a generative model is feasible. CoRe-TFM instead targets the increasingly common case in which a high-performing frozen discriminative TFM already exists and multiple conditional queries are assembled downstream.

9 Limitations and Threats to Validity

First, the current formulation is pairwise and categorical. Global consistency across many targets introduces additional compatibility constraints and potentially exponential state spaces. Second, regression and mixed discrete–continuous targets are deferred because continuous predictive heads require an explicit density representation or controlled discretization. Third, the five-fold TabICLv2 Car pilot establishes released-model feasibility but covers only one model–data set cell; exact-view and surrogate experiments therefore still cannot replace the planned broader released-model validation. Fourth, validation selection consumes data and can overfit in small samples; the validation-size study quantifies but does not eliminate this limitation. Fifth, ECE is only descriptive; conclusions rely primarily on proper scores and exact truth distances where available. Sixth, the TFM literature is moving rapidly, so the novelty audit must be repeated immediately before submission. Finally, the 60-task mixture benchmark is a designed stress test of view reliability, not a claim about the empirical distribution of errors produced by any specific TFM.

10 Reproducibility

The implementation, tests, synthetic generators, perturbation studies, selection experiments, pinned data specifications, reconciliation solvers, and plotting scripts are maintained in the project repository. All adaptive choices are separated from test scoring. Random seeds are explicit. The repository now includes fold-level output for the five-fold TabICLv2 Car pilot. The complete real-data release will additionally archive raw data snapshots by source identifier and hash, model package versions, checkpoint identifiers, license/access conditions, hardware, inference configuration, and scripts that regenerate every table and figure.

11 Conclusion

CoRe-TFM reframes probability consistency in tabular foundation models as a four-view reliability problem rather than a projection problem with a universal answer. Hard CoRe is appropriate when direct marginals are trustworthy; Soft CoRe makes marginal trust continuous; and Selective CoRe allows validation evidence to reject reconciliation entirely. Controlled experiments show that hard preservation helps exactly when its trust assumption is satisfied, fails when that assumption is violated, and cannot outperform a genuinely reliable chain direction merely by enforcing coherence. In heterogeneous tasks, Selective CoRe significantly improves over the best fixed policy and approaches the oracle candidate family as validation evidence grows. A five-fold TabICLv2 Car pilot adds an important real-model counterexample: incompatibility is substantial, but the best raw chain direction still beats pooling and reconciliation in joint log loss. The remaining decisive step is the full pre-specified benchmark across released TFMs and data sets. Only after that matrix is complete should the paper claim average gains for deployed foundation models.

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Appendix A. Additional Derivations

A.1 Why arithmetic and geometric pools are KL barycenters

For normalized distributions P_1 and P_2, minimizing $w\,\mathrm{KL}(P_1\|Q)+(1-w)\,\mathrm{KL}(P_2\|Q)$ over normalized Q is equivalent to maximizing $\sum_z[wP_1(z)+(1-w)P_2(z)]\log Q(z)$, giving the arithmetic pool. Reversing the KL arguments yields an objective whose Q-dependent part is $\sum_z Q(z)\log Q(z)-\sum_z Q(z)[w\log P_1(z)+(1-w)\log P_2(z)]$, whose normalized minimizer is the weighted geometric pool.

A.2 Four-view decomposition

For $J^{B\to A}(a,b)=p_B(b)p_{A|B}(a\mid b)$,

$$\mathrm{KL}(Q\|J^{B\to A})=\sum_{a,b}Q(a,b)\log\frac{Q_B(b)Q_{A|B}(a\mid b)}{p_B(b)p_{A|B}(a\mid b)} \tag{18}$$

$$=\mathrm{KL}(Q_B\|p_B)+\mathbb{E}_{b\sim Q_B}\,\mathrm{KL}(Q_{A|B=b}\|p_{A|B=b}). \tag{19}$$

The $J^{A\to B}$ decomposition is symmetric. This identity is what makes Soft CoRe a four-view rather than two-matrix objective.

Appendix B. Exact-View Perturbation Details

For each exact probability vector p, perturbation is performed in log space,

$$\tilde{p}_i=\frac{\exp(\log p_i+\sigma\epsilon_i)}{\sum_j\exp(\log p_j+\sigma\epsilon_j)},\qquad \epsilon_i\sim\mathcal{N}(0,1). \tag{20}$$

The four view-specific σ values define the reliability regime. This preserves positivity and normalization while allowing marginal and conditional errors to be controlled independently.

The prespecified paired comparisons per regime are Hard CoRe versus arithmetic, Adaptive Soft versus arithmetic, Adaptive Soft versus Hard CoRe, and Adaptive Soft versus validation-selected original order. Holm correction is applied within each regime. The most important sign reversal is Hard minus arithmetic: -0.013799 when marginals are reliable, but $+0.142265$ when conditionals are reliable and marginals are noisy; both adjusted p-values are 8×10^{-6}.

Appendix C. Selective Candidate Family

The selector includes two original orders, arithmetic pooling, weighted geometric pooling, weighted Hard CoRe, and weighted Soft CoRe. Direction weights are $\{0,0.25,0.5,0.75,1\}$ and soft marginal penalties are $\{0.03,0.1,0.3,1,3,10,30\}$. Duplicate half-weight geometric and hard candidates are removed, leaving 48 candidates. Ties prefer lower-complexity policies and less-extreme hyperparameters. The full-family oracle in Sec. 7.3 evaluates exactly this same set using test truth and is never used for deployment or hyperparameter selection.

Appendix D. Real-TFM Reporting Checklist

For each model–data set cell, the final release should report: factorization TV; both marginalization TVs; joint NLL/Brier; both induced conditional NLL/Brier scores; marginal NLL/Brier/accuracy; calibration summaries; reconciliation and marginal distortion; inference and reconciliation runtime; selected policy/hyperparameters; fold-level values; package/checkpoint identifiers; and hardware. Cross-model conclusions should be based on paired data set–fold summaries with multiplicity-aware testing or hierarchical analysis, not on individual rows treated as independent observations.

References

Jean-David Benamou, Guillaume Carlier, Marco Cuturi, Luca Nenna, and Gabriel Peyre. Iterative bregman projections for regularized transportation problems. *SIAM Journal on Scientific Computing*, 37(2):A1111–A1138, 2015. doi: 10.1137/141000439.

Marko Bohanec. Car evaluation. UCI Machine Learning Repository, 1988.

Lenaic Chizat, Gabriel Peyre, Bernhard Schmitzer, and Francois-Xavier Vialard. Scaling algorithms for unbalanced transport problems. *Mathematics of Computation*, 87(314): 2563–2609, 2018. doi: 10.1090/mcom/3303.

Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, 2013.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

Google Research. Introducing tabfm: A zero-shot foundation model for tabular data, 2026. Research release, accessed August 2026.

Leo Grinsztajn, Klemens Floege, Oscar Key, Felix Birkel, Philipp Jund, Brendan Roof, et al. Tabpfn-3: Technical report. *arXiv preprint arXiv:2605.13986*, 2026. doi: 10.48550/arXiv.2605.13986.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330, 2017.

Noah Hollmann, Samuel Mueller, Lennart Purucker, Arjun Krishnakumar, Max Koerfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637:319–326, 2025. doi: 10.1038/s41586-024-08328-6.

Rafael Izbicki and Pedro L. C. Rodrigues. Benchmarking tabular foundation models for conditional density estimation in regression. *arXiv preprint arXiv:2603.26611*, 2026.

Tyler R. Johnson, Kian Ben-Jacob, Nima Negarandeh, Oriol Vendrell-Gallart, and Ramin Bostanabad. On the uncertainty quantification ability of tabular foundation models. *arXiv preprint arXiv:2606.01427*, 2026.

Christian Kloetergens, Vijaya Krishna Yalavarthi, Lars Schmidt-Thieme, and Tom Hanika. Do tabular foundation models agree with themselves? *arXiv preprint arXiv:2608.06004*, 2026. doi: 10.48550/arXiv.2608.06004.

Jonas Landsgesell and Pascal Knoll. Scoringbench: A benchmark for evaluating tabular foundation models with proper scoring rules. *arXiv preprint arXiv:2603.29928*, 2026. doi: 10.48550/arXiv.2603.29928.

Seunghan Lee, Jaehoon Lee, Jun Seo, Sungdong Yoo, Minjae Kim, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, SoonYoung Lee, and Wonbin Ahn. Mitigating label shift in tabular in-context learning via test-time posterior adjustment. *arXiv preprint arXiv:2605.04363*, 2026. doi: 10.48550/arXiv.2605.04363.

Mahbod Majid, Rattana Pukdee, Vishwajeet Agrawal, Burak Varici, and Pradeep Kumar Ravikumar. On the consistent recovery of joint distributions from conditionals. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics*, volume 258 of *Proceedings of Machine Learning Research*, pages 4708–4716, 2025.

Joseph Mure. Optimal compromise between incompatible conditional probability distributions, with application to objective bayesian kriging. *arXiv preprint arXiv:1703.07233*, 2017. doi: 10.48550/arXiv.1703.07233.

Jingang Qu, David Holzmueller, Gael Varoquaux, and Marine Le Morvan. Tabiclv2: A better, faster, scalable, and open tabular foundation model. *arXiv preprint arXiv:2602.11139*, 2026. doi: 10.48550/arXiv.2602.11139.

Juntong Shi, Minkai Xu, Harper Hua, Hengrui Zhang, Stefano Ermon, and Jure Leskovec. Tabdiff: A mixed-type diffusion model for tabular data generation. In *International Conference on Learning Representations*, 2025.

Robin Young. On the conditioning consistency gap in conditional neural processes. *arXiv preprint arXiv:2604.19312*, 2026. doi: 10.48550/arXiv.2604.19312.

Hengrui Zhang, Liancheng Fang, Qitian Wu, and Philip S. Yu. Tabnat: A continuous-discrete joint generative framework for tabular data. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 74858–74881, 2025.